



A
PROJECT REPORT
ON
“Startup Funding & Trend Analysis”

SUBMITTED BY
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UNDER GUIDANCE
“Mr.Rushi Durge”

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Pimpri, Pune-18

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I would like to thank my teacher Mr.Rushi Durge, who gave me a golden opportunity to work on this project.

I would also like to thank my parents and friends for their immense support and encouragement during this project. Without their help, completing this project would have been very difficult.

Your's sincere,
Vivek Jadhav

(T.Y.BSC.CS)

DECLARATION

We undersigned **“Vivek Jadhav”** hereby declare that the project work entitled **“Startup Funding & Trend Analysis”** is conducted under the guidance of **“Mr.Rushi Durge”**. It is my original work and has not been copied from any other source or project work submitted to the University.

Date:

(“Vivek Jadhav”)



**College Of Commerce, Science and Information Technology,
Pimpri, Pune-18**

CERTIFICATE

This is to certify that **Vivek Jadhav** are students of TY B.Sc.(CS) SEM-VI course have satisfactorily completed the project report entitled as **Startup Funding & Trend Analysis** in the partial fulfilment of the Bachelor of Science (Computer Science) course during the academic year 2025-2026.

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1.Introduction

This project focuses on analyzing startup funding data to understand investment trends across different years, cities, industries, and funding rounds. The main aim of the project is to identify how startup funding has evolved over time and what factors influence higher funding amounts. The analysis is performed using Python in Jupyter Notebook for data cleaning, preprocessing, and exploratory data analysis. In addition to this, an interactive dashboard is created in Power BI to visually represent the insights through dynamic charts, KPIs, filters, and drill-down features. The combination of Python and Power BI helps in understanding which cities attract the highest funding, which industries show rapid growth, and how investor behaviour changes across years. Overall, this project highlights the key trends and patterns present in the Indian startup ecosystem and provides a clear, data-driven view of startup funding activity.

1.1 Motivation

The motivation behind choosing this project comes from the rapidly growing startup ecosystem in India. Startups have become an important part of economic development, and understanding how funding trends change over the years can provide valuable insights. I wanted to explore which cities and industries attract the most investment and how funding patterns shift over time. This project also motivated me to learn practical data analysis skills, use Python for real-world data, and create an interactive dashboard using Power BI to visualize meaningful insights. Overall, the desire to understand real startup trends and improve my data analysis abilities inspired me to work on this project.

1.2 Problem statement :

The Indian startup ecosystem is growing rapidly, but funding patterns are not clearly understood. There is no simple way to identify which cities receive the highest funding, which industries attract investors, how funding amounts change across years, and what factors affect investment decisions. The data is large, unorganized, and difficult to analyze without proper tools. This lack of clarity makes it hard for stakeholders to understand market trends and make informed decisions.

This project aims to solve this problem by analyzing startup funding data using Python and presenting key insights through an interactive Power BI dashboard. The goal is to provide a clear understanding of funding trends, top-performing cities, industries, funding rounds, and yearly investment patterns.

1.3 Purpose

The purpose of this project is to analyze startup funding data to identify important trends, patterns, and insights related to investment activities in India. By using Python for data analysis and Power BI for interactive visualization, the project aims to make the data easy to understand for users and decision-makers. The goal is to provide clear information about which cities, industries, and funding rounds receive the most investment, how funding has changed over time, and what factors may influence higher funding amounts. Ultimately, the purpose is to support better understanding and decision-making based on real data.

• Objectives :

Objectives of the Project :

To analyze startup funding trends across different years.

To identify top cities that receive the highest funding.

To examine funding patterns across various industries/verticals.

To study investor participation and funding rounds.

To clean, preprocess, and analyze the dataset using Python in Jupyter Notebook.

To create interactive dashboards in Power BI to visually represent insights and allow dynamic filtering.

To understand key factors that influence higher funding amounts.

Goals :

To gain a deep understanding of the Indian startup funding ecosystem using real-world data.

To identify long-term patterns and trends in funding across cities, industries, years, and rounds.

To build strong data analysis skills through Python-based cleaning, analysis, and visualization.

To create a professional interactive dashboard in Power BI that helps users explore funding insights easily.

To support informed decision-making for stakeholders by presenting clear and meaningful data insights.

To provide a comprehensive overview of investment behaviour and highlight areas of high growth potential.

1.4 Literature survey :

Several studies and industry reports have examined the growth of the Indian startup ecosystem and the funding patterns across different sectors. Research from NASSCOM, Startup India, and consulting firms such as KPMG and PwC highlights that cities like Bengaluru, Mumbai, and Delhi NCR consistently receive the highest startup funding due to strong investor networks and technological infrastructure. Many studies also show that sectors like FinTech, EdTech, E-commerce, and HealthTech attract significant investment because of high market demand and innovation opportunities.

Previous research commonly uses data analytics tools such as Python, R, and Excel to analyze funding distribution, investor behaviour, and year-wise investment trends. However, most of these studies present static graphs and limited interactive visualizations. They also often focus on specific segments rather than providing a complete, combined view of cities, industries, funding rounds, and investors.

This project builds upon previous work by using Python for detailed exploratory data analysis and Power BI for interactive dashboard creation. It fills the gap by offering a more comprehensive, user-friendly, and visually interactive representation of startup funding trends across India, making the insights easier to interpret for researchers, students, and decision-makers.

1.5 Scope and Limitation :

- **The scope of this project** includes the complete analysis of startup funding data to understand investment patterns across different years, cities, industries, and funding rounds. The project covers data cleaning, preprocessing, and exploratory analysis using Python in Jupyter Notebook. It also includes the development of an interactive Power BI dashboard to visualize key insights through charts, KPIs, and filters.

The project focuses on identifying trends in funding amounts, top-funded cities, investor involvement, industry-wise funding distribution, and yearly growth or decline in investments. The analysis is limited to the available dataset and does not include external market or economic factors. Overall, the project provides a comprehensive view of the Indian startup funding landscape based on the given data.

- **Limitations of the Project**

The analysis is based only on the provided dataset and does not include external market factors or international startup data.

Some data entries may contain missing or inaccurate values, which can affect the results.

The project does not perform predictive modelling or machine learning; it focuses only on descriptive data analysis.

Funding data may not be complete for all years, and some startups may have incomplete information.

The Power BI dashboard depends on manual data updates and does not support real-time data refresh in this project.

Industry names, investor names, and city names may vary due to inconsistent formatting in the dataset.

2. System Analysis

The system used in this project processes startup funding data through a series of structured steps to generate meaningful insights. The system begins with raw data input, which includes information such as startup names, funding amounts, cities, industries, investors, and funding years.

Once the data is collected, it undergoes cleaning and preprocessing to remove missing values, duplicates, and formatting errors. This processed data is then analyzed using Python to identify patterns, trends, and relationships. Exploratory Data Analysis (EDA) is performed to understand year-wise funding changes, city-wise distribution, industry performance, and investor activity.

The system further includes a visualization layer, where Python and Power BI are used to create charts, graphs, and an interactive dashboard. These visualizations help users explore the data dynamically using filters and slicers. Finally, the system outputs clear insights, KPIs, and visual representations that help users understand the Indian startup funding ecosystem effectively.

• Components of Project :

Dataset

Python (Jupyter Notebook)

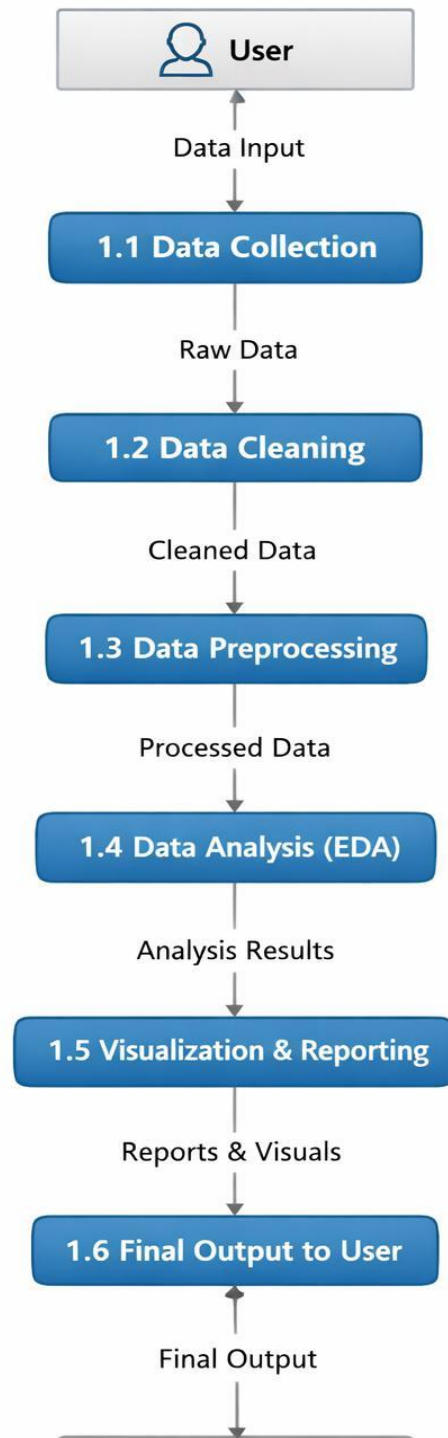
Libraries (Pandas, NumPy, Matplotlib, Seaborn)

Power BI

Final Dashboard

User

- **Data Flow Diagram :**



Level 1 Data Flow Diagram

2.1 Existing systems

Currently, startup funding data is analyzed using traditional methods such as Excel sheets, basic reporting tools, or manual analysis. These methods involve manually checking the funding amount, city-wise data, investor information, and industry trends. Many existing analyses are not automated and lack interactive dashboards. Users need to check multiple files, documents, or websites to understand funding patterns across different years and industries.

2.2 Scope and Limitations of existing systems :

• scope

The existing system (manual analysis, Excel reports, basic tools) has a limited but useful scope :

It allows storing startup funding data in tables or spreadsheets.

Users can calculate totals, averages, and basic statistics manually.

Basic charts like bar graphs or pie charts can be created using Excel.

Data can be filtered manually for specific cities, industries, or years.

Reports can be prepared, but without automation or interactivity.

• Limitations

No automation

Every update or calculation must be done manually.

Time-consuming and prone to errors.

Poor visualization

No interactive dashboards

Cannot drill down into deeper insights

Difficult to compare trends

Hard to compare multiple years, industries, or cities

No dynamic filtering

Data cleaning is manual

Removing missing values, duplicates, and formatting is tedious.

Limited analytical capability
Cannot detect patterns, trends, or anomalies easily
No advanced charts or analysis
Scalability issues
Excel struggles with large datasets
Slow performance
No integration with advanced tools
Cannot connect with real-time dashboards
No integration with Python or BI tools
Reporting is static
Once created, reports do not update automatically
Every change requires re-doing the work

2.3 Project Perspective and features

• Project Perspective

This project is designed to provide a more efficient, interactive, and data-driven approach to understanding startup funding trends in India. Unlike traditional systems that rely on manual analysis and static reports, this project integrates Python-based data cleaning and analysis with Power BI's interactive dashboard capabilities. The system offers a unified platform where users can explore funding patterns across years, cities, industries, and investors in a dynamic manner.

The project serves:

Students and researchers studying startup trends
Investors who want to understand market patterns
Entrepreneurs exploring funding insights
Data analysts who require interactive visualizations

• Project Features

1. Data Cleaning and Preprocessing

Removal of duplicates

Handling missing values

Standardizing formats

Converting data types (e.g., funding amount)

2. Exploratory Data Analysis (Python)

Year-wise funding trends

City-wise comparison

Industry/vertical-wise growth

Investor-level analysis

3. Power BI Interactive Dashboard

Dynamic filters for year, city, industry, funding round, investor

KPIs such as:

Total funding

Average funding

Total startups

Drill-through and tooltip analysis

Slicers for quick navigation

Attractive visual charts (bar, line, map, donut)

4. Startup Ecosystem Insights

Identification of top-funded cities

Fast-growing industries

Years with highest funding

Investor behavior and patterns

5. User-Friendly Interface

Easy navigation using Power BI buttons

Clear visual layout

Real-time dynamic updates based on slicer selection

6. Data Reporting

Summaries and charts exportable as PDF/image

Ready-to-present visuals for academic or professional use.

3. System Design

1. System Architecture

The system uses a two-layer architecture:

a) Backend (Python – Jupyter Notebook)

Data collection

Data cleaning

Data preprocessing

Exploratory Data Analysis (EDA)

Exporting cleaned data for Power BI

b) Frontend (Power BI Dashboard)

Visualization

Interactive filtering

KPI generation

User interface

This combination forms an end-to-end analytical system.

2. System Components

Your project consists of the following major components:

a) Data Input Module

Loads dataset from CSV/Excel

Reads into Python using Pandas

b) Data Cleaning Module

Remove duplicates

Handle missing values

Convert data types

Standardize formats

c) Data Processing Module

Create new columns if needed

Grouping & aggregation (year, city, industry)

Summarization

Export cleaned data

d) Analysis Module (Python)

Year-wise trends

City-wise insights

Industry-wise performance

Funding distribution

Basic charts

e) Visualization Module (Power BI)

Import cleaned CSV

Create charts (bar, line, map, donut)

Define KPIs

Add slicers & filters

Build interactive dashboard

f) Output Module

Final Power BI dashboard

Insights for decision-making

3. Technology Stack

Python (Pandas, NumPy, Matplotlib, Seaborn)

Jupyter Notebook

Power BI (for dashboard)

4. Workflow of the System

Step-by-step system functioning:

- Load raw dataset
- Clean and preprocess using Python
- Perform EDA to understand basic patterns
- Export cleaned data

- Import into Power BI
- Build visualizations
- Add filters and KPIs
- Combine visuals into an interactive dashboard
- User views insights

3.1 Design Constraints

1. Technical Constraints

The project is built using only Python (Jupyter Notebook) and Power BI, so advanced big-data tools like Spark or Hadoop cannot be used.

Power BI has limitations with extremely large datasets, which restricts scalability.

Python visualization is limited to Matplotlib/Seaborn since the project must remain simple.

2. Data Constraints

Dataset contains missing values and duplicates, which affect the accuracy of insights.

Some cities, industries, or years have very low data, limiting deep analysis.

Data may be outdated or incomplete, affecting current trend interpretation.

3. Hardware Constraints

The project was executed on a normal laptop, limiting:

Processing speed

Memory usage

Ability to handle large datasets

High-volume Power BI models may load slowly.

4. Time Constraints

Limited time to complete the project restricted the use of:

Advanced machine learning models

Real-time dashboards

Predictive analysis

5. Skill Constraints

The project is developed using beginner/intermediate-level Python and Power BI skills.

Deep learning, advanced forecasting, and automation were not included.

6. Software Constraints

Only free/academic versions of tools were used.

Limited access to premium Power BI features (no Power BI Pro).

No integration with APIs or live databases.

7. Visualization Constraints

Power BI allows only certain chart types.

Too many visuals reduce dashboard performance.

Real-time updates were not possible due to static dataset.

3.2 System Model using Oose

- **case diagram**



Use Case Diagram for Startup Funding Analysis System

- **Sequence flow diagram**

Process Flow

User → DataLoader: uploadDataset()

The user uploads the startup funding CSV file.

DataLoader → DataLoader: read dataset

The system loads the dataset into memory.

DataLoader → DataPreprocessor: send dataset

The dataset is passed for cleaning and preprocessing.

DataPreprocessor → DataPreprocessor: cleanData()

Missing values, formatting, and data types are corrected.

DataPreprocessor → Analyzer: send preprocessed data

Cleaned data is forwarded for analysis.

Analyzer → Analyzer: performAnalysis()

The Analyzer calculates:

Funding trends

City-wise funding

Industry growth

Investor activity

Analyzer → VisualizationGenerator: send analysis output

The results are sent for chart creation.

VisualizationGenerator → VisualizationGenerator: generateCharts()

Generates:

Bar charts

Line charts

Pie charts

KPI cards

VisualizationGenerator → Dashboard: send charts

Charts and KPIs are sent to the dashboard.

Dashboard → User: displayDashboard()

User views interactive insights.

3.3 Data Model

- **Introduction to data model**

The data model defines how startup-related information is organized in the project.

It represents the entities, attributes, and relationships that describe the Indian startup funding ecosystem.

The model helps the system perform analysis such as year-wise trends, city-wise comparisons, industry insights, and investor patterns.

- **entities and attributes**

Entity 1

Entity 1: Startup	
Attribute	Description
startup_name	Name of the startup
city	Location of startup (e.g., Bangalore, Mumbai)
state	State where the startup is based
industry / vertical	Industry category of the startup
sub-vertical	More detailed category
founded_year	Year startup was founded

Entity 2

Entity 2: Funding	
Attribute	Description
amoun_crores	Funding amount_received (in Cr)
round	Series of funding (Seed/Series A/Series B etc.)
date	Funding date
year	Extracted year from the date
investor	(If present) Startup valuation post-funding

Entity 3

Entity 3: Investor	
Attribute	Description
investor_name	Name of investor
investor_type	VC, Angel, Private Equity (if available)

• Cleaned Dataset Structure (After Python Preprocessing)

Startup_name

City

State

Industry

Year

Amount_crores

Round

Investor

This cleaned model ensures consistent formats and reliable analysis.

• Data Relationships in Power BI

If needed, include this:

Since the project uses a single flat-file dataset, Power BI works with a single-table model, but logical relationships exist:

Startup → Funding (1-to-many)

One startup may receive multiple funding rounds.

Year → Funding (1-to-many)

One year contains many funding events.

City → Startup → Funding

A city has many startups, and each startup can have multiple fundings.

• Logical Data Model

Startup (startup_name, city, state, industry)

|

| 1-to-many

|

Funding (startup_name, amount_crores, investor, round, date, year)

|

| many-to-1

|

Investor (investor_name)

- **Purpose of Data Model**

Helps ensure data consistency

Organizes complex startup information

Simplifies filtering (city, year, industry, investor)

Supports Power BI dashboards

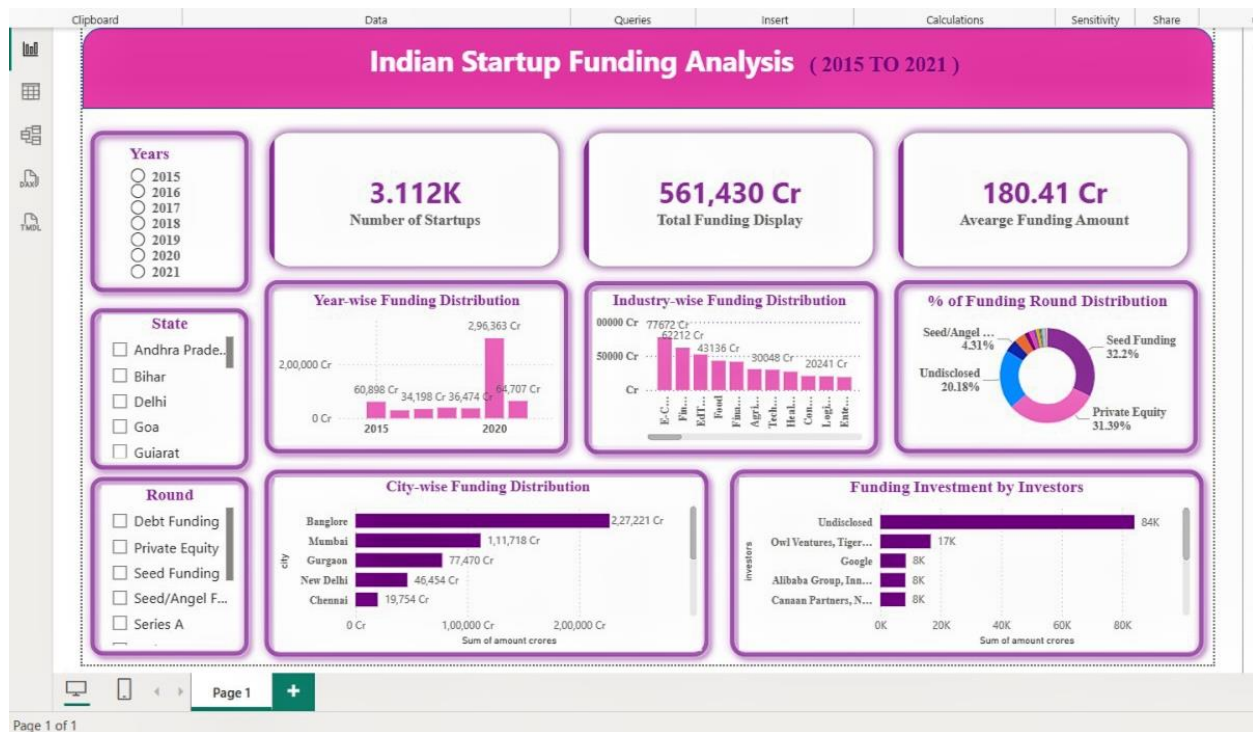
Enables accurate trend analysis

3.4. User Interfaces

The User Interface of the Startup Funding Analysis System is designed to provide users with a clean, interactive, and visually rich experience. Power BI dashboards and Jupyter Notebook visual charts help users explore funding trends across years, cities, and industries.

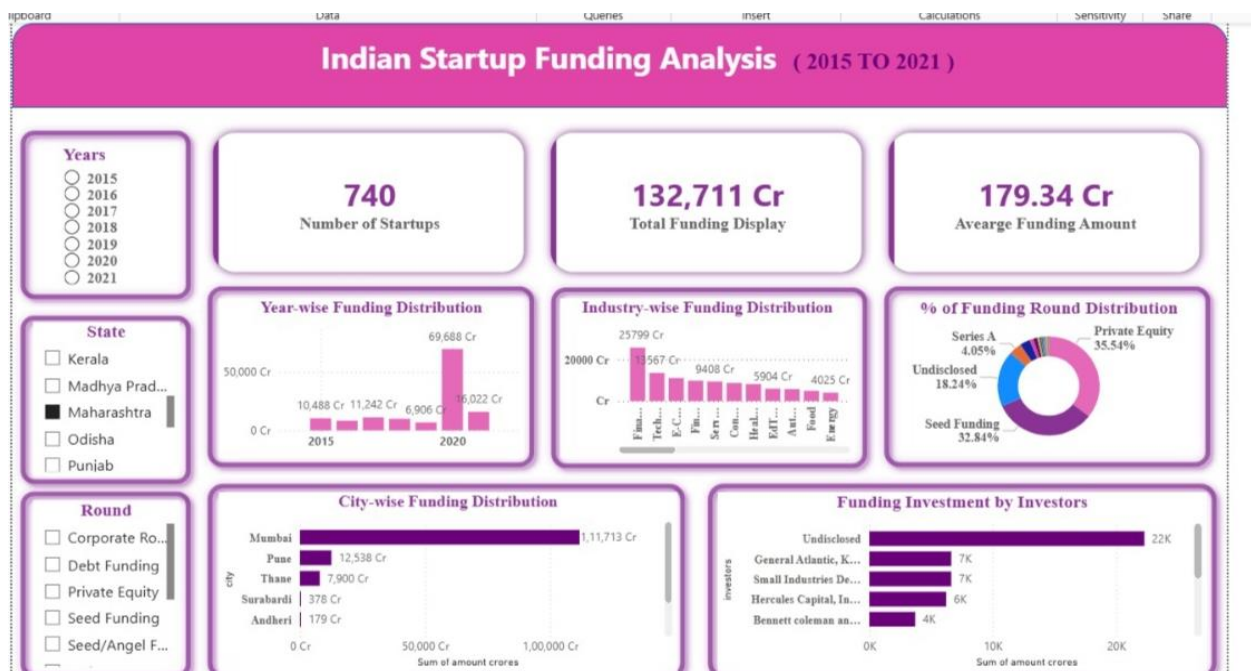
- **Main Dashboard Interface (Power BI)**

The main dashboard displays key performance indicators (KPIs) such as Total Funding, Number of Startups, and Average Funding Amount. It also includes year-wise slicers that allow users to filter the data instantly.



• State – wise Visualizations

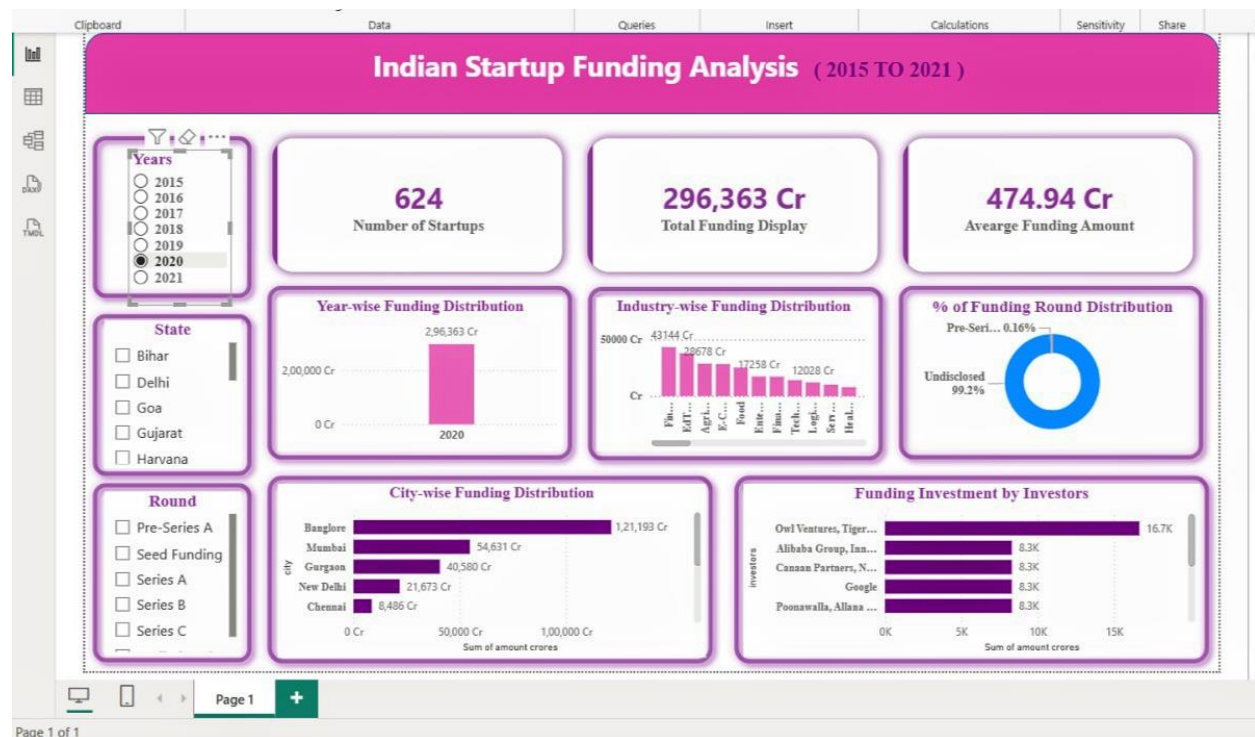
This section of the dashboard provides bar charts/pie charts that show funding distribution across different States. Users can click on any category to drill down into specific insights.



• Interactive Slicer Functionality

The dashboard includes a Year Slicer that allows the user to filter the entire dataset by selecting a specific year. When a particular year (e.g., 2018) is selected, all the dashboard visuals such as KPIs, bar charts, and industry-wise distributions update instantly to show insights only for that selected year.

This interactive behavior helps users quickly analyze how startup funding trends changed over different years.



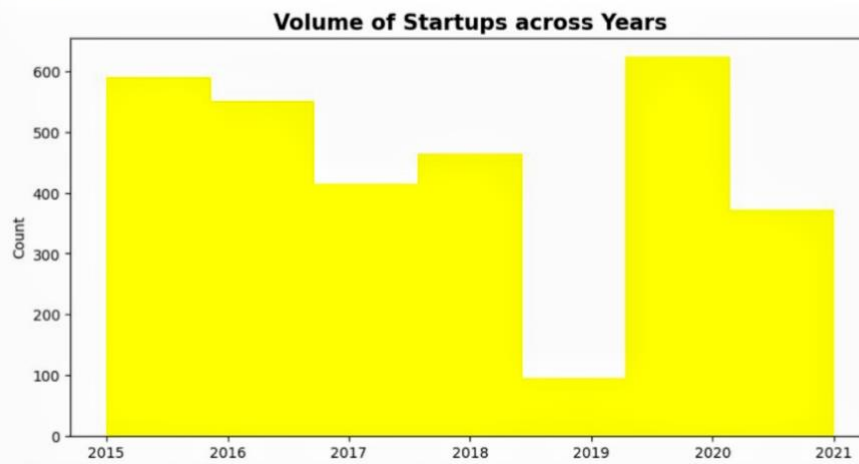
• Jupyter Notebook Visualization

In addition to the Power BI dashboard, an initial exploratory visualization was created in Jupyter Notebook to understand the distribution of startup funding data across different years. A histogram was plotted to show how many startups received funding in each year.

This visualization helped identify trends such as peak funding years, variations in startup activity, and overall data spread before building the final interactive dashboard.

```
[96]: #DISTRIBUTION OF STARTUPS YEAR TO YEAR (HISTOGRAM CHART)
```

```
plt.figure(figsize=(10,5))
plt.hist(df['year'],
bins=len(df['year'].unique()), color='yellow')
plt.title("Volume of Startups across Years",fontsize=15, fontweights='bold')
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```



4. Implementation Details

The implementation phase of this project involves the practical execution of all tasks required to analyze the dataset, extract insights, and develop an interactive dashboard. This project integrates multiple tools such as Python, Jupyter Notebook, and Power BI to ensure efficient data processing and clear visualization of results. The implementation process includes data collection, data cleaning, feature engineering, visualization, and dashboard creation. Each step is performed systematically to ensure accuracy, usability, and reliability of the final output.

The purpose of this section is to document all the requirements—both software and hardware—used during the development of the project. This ensures that the project environment can be recreated or extended by others in the future.

4.1 Software Specifications

The software specifications include all the tools, programming languages, libraries, and platforms used to perform data analysis, create machine learning models (if applicable), and design the final dashboard. These software components ensure smooth execution and provide the capability to handle structured as well as unstructured data.

Detailed Software Requirements :

- **Operating System: Windows 10/11**

The entire project was implemented on a Windows operating system due to its compatibility with Power BI Desktop, Jupyter Notebook, and other required tools. Windows provides a stable environment for running Python scripts, performing data preprocessing, and creating visual dashboards.

- **Programming Language: Python 3.x**

Python is used for loading the dataset, cleaning data, performing exploratory data analysis (EDA), and developing models. It offers extensive libraries that simplify complex operations and allow automation of tasks.

- **Jupyter Notebook / VS Code**

Jupyter Notebook is used as the primary interface for writing and executing Python code. It provides an interactive environment where results can be visualized instantly. It is ideal for step-by-step data analysis and documentation.

- **Python Libraries**

Pandas: For data cleaning and manipulation

NumPy: For numerical operations

Matplotlib & Seaborn: For data visualization

Scikit-learn: For predictive modeling (if used)

- **Power BI Desktop (Latest Version)**

Power BI is used to create interactive visualizations and dashboards. It supports drag-and-drop visual creation, DAX measures, slicers, and filters. The final dashboard is prepared in Power BI to present insights clearly.

- **Web Browser (Chrome/Edge)**

A browser is required for accessing online documentation, downloading datasets, and verifying results.

- **Dataset Format: CSV/Excel**

The project dataset is stored in CSV/Excel format, which is easily readable by both Python and Power BI.

4.2 Hardware Specifications

Hardware specifications describe the minimum and recommended physical components required to run the project smoothly. Since this is a data analysis project involving large datasets, having a capable machine ensures faster execution and lag-free visualization.

Detailed Hardware Requirements :

- **Processor (CPU): Intel Core i3 / AMD Ryzen 3 or Higher**

A multi-core processor ensures the system can handle Python scripts, Power BI operations, and visualization rendering without freezing. Higher-end processors (i5/i7/Ryzen 5 or above) provide faster performance during large computations.

- **Memory (RAM): Minimum 4 GB, Recommended 8 GB or More**

RAM plays a major role in loading datasets, running Jupyter Notebook, and working with Power BI dashboards. Although 4 GB is enough for basic execution, having 8 GB or more significantly improves speed and reduces system lag.

- **Storage: Minimum 10–20 GB Free Space**

Storage is required for:

Dataset files

Jupyter notebooks

Power BI (.pbix) files

Exported reports and visuals

SSD storage is recommended for faster file access and better overall system performance.

- **Graphics Capability**

Integrated graphics are sufficient for this project as visualization is handled by software like Power BI. No dedicated GPU is required unless the dataset is extremely large.

- **Monitor / Display**

A minimum screen size of 14 inches is required for comfortable viewing of dashboards and code. Larger screens (15.6" or external monitor) improve readability and help during presentations.

- **Input Devices: Keyboard and Mouse**

Essential for interacting with development environments, writing code, and navigating dashboard elements.

- **Internet Connectivity**

Required for downloading libraries, accessing documentation, and obtaining datasets.

5. Outputs and Reports

This section presents all the outputs generated by the system during the development of the Startup Funding Analysis project. It includes backend results from data import, cleaning, preprocessing, and machine learning execution. It also includes the testing strategy and validation results used to ensure the correctness, accuracy, and functionality of the entire system.

The purpose of this chapter is to provide a clear understanding of how the system processes the raw dataset, transforms it into a usable form, and finally produces meaningful insights through analytical and predictive methods. Each output shown in this section represents an essential step of the project workflow.

5.1 System Outputs

System outputs are the results directly generated by the system during execution in Jupyter Notebook. These outputs include the loading of the dataset, the cleaning process, intermediate tables, and machine learning model results. These outputs are important because they verify that the underlying logic and implementation are functioning correctly

- **Data Import Output**

The first step in the analysis process involved importing the dataset into the Jupyter Notebook environment using the Pandas library. After loading the dataset, the first few rows were displayed to ensure that all columns—such as Startup Name, Industry, City, State, Funding Amount (Crores), and Year—were correctly imported.

This output confirms that the dataset is successfully read and no issues occurred during loading such as missing columns, encoding errors, or incorrect file format.

```
Jupyter ind_funding Last Checkpoint: 2 days ago
File Edit View Run Kernel Settings Help
JupyterLab Python [conda env:base]

[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
from sklearn.linear_model import LinearRegression

[2]: df = pd.read_csv(r'C:\Users\prati\Downloads\archive\vivek\indian_startups_funding.csv', encoding='unicode_escape')

[5]: df.head(10)

[5]:
```

	Unnamed: 0	Sr No	startup	vertical	city	investors	round	amount	year	state
0	0	1	#Fame	Media	Mumbai	To the new ventures	Private Equity	82.50	2015	Maharashtra
1	1	2	1mg	Healthcare	Gurugram	Deep kalra, Sequoia capital, Omidyar network, ...	Private Equity	49.50	2015	Haryana
2	2	3	20Dresses	Personal Care	Mumbai	Srinivas aluri, Srinivas modukury, Bs nagesh	Seed Funding	8.25	2015	Maharashtra
3	3	4	33Coupons	Marketing	Kanpur	Not disclosed	Seed Funding	0.17	2015	Uttar Pradesh
4	4	5	6Degree	Networking	Mumbai	Centre for incubation & business acceleration ...	Seed Funding	0.00	2015	Maharashtra
5	5	6	aagaar	Food	Gurugram	Group of investors	Seed Funding	0.00	2015	Haryana
6	6	7	Aasaanjobs	Careers	Mumbai	Inventus capital partners, Idg ventures	Seed Funding	12.38	2015	Maharashtra
7	7	8	Abhibus	Travel	Hyderabad	Paytm (one97 communications)	Private Equity	0.00	2015	Telangana
8	8	9	Absentia	Technology	Bengaluru	Astarc ventures, 50k ventures & others	Seed Funding	1.49	2015	Karnataka
9	9	10	Actionable Analytics	Data Analytics	New Delhi	B m gupta, Vinod singhal	Seed Funding	0.00	2015	Delhi

```
[7]: df.shape
```

• Data Cleaning Output – Step 1

The dataset initially contained missing values, duplicated rows, inconsistent text formatting, and unstructured numerical columns.

In the first stage of cleaning:

Null values were identified and handled

Duplicate entries were removed

Column names were standardized

Data types were converted where necessary

The intermediate output displayed the cleaned dataframe and verified that the unwanted records were removed. This step is crucial for ensuring accurate analysis and avoiding misleading results later.

```
[25]: df.columns

[25]: Index(['startup', 'vertical', 'city', 'investors', 'round', 'amount', 'year',
        'state'],
        dtype='object')

[27]: df['city'] = df['city'].replace({'Bengaluru': 'Bangalore', 'Gurugram': 'Gurgaon'})

[29]: df.head(10)

[29]:
```

	startup	vertical	city	investors	round	amount	year	state
0	#Fame	Media	Mumbai	To the new ventures	Private Equity	82.50	2015	Maharashtra
1	1mg	Healthcare	Gurgaon	Deep kalra, Sequoia capital, Omidyar network, ...	Private Equity	49.50	2015	Haryana
2	20Dresses	Personal Care	Mumbai	Srinivas aluri, Srinivas modukury, Bs nagesh	Seed Funding	8.25	2015	Maharashtra
3	33Coupons	Marketing	Kanpur	Undisclosed	Seed Funding	0.17	2015	Uttar Pradesh
4	6Degree	Networking	Mumbai	Centre for incubation & business acceleration ...	Seed Funding	0.00	2015	Maharashtra
5	aagaar	Food	Gurgaon	Group of investors	Seed Funding	0.00	2015	Haryana
6	Assaanjobs	Careers	Mumbai	Inventus capital partners, Idg ventures	Seed Funding	12.38	2015	Maharashtra
7	Abhibus	Travel	Hyderabad	Paytm (one97 communications)	Private Equity	0.00	2015	Telangana
8	Absentia	Technology	Bangalore	Astarc ventures, 50k ventures & others	Seed Funding	1.49	2015	Karnataka
9	Actionable Analytics	Data Analytics	New Delhi	B m gupta, Vinod singhal	Seed Funding	0.00	2015	Delhi

```

[31]: df.dropna(subset=['investors'], inplace=True)

[33]: df.shape

[33]: (5070, 8)

```

```

[17]: # 1 WORD FOR MANY ex., 'Undisclosed'='Undisclosed', 'Undisclosed investors', 'Undisclosed investor', 'Not disclosed', 'Undisclosed'

df['investors'] = df['investors'].replace({
    'Undisclosed': 'Undisclosed',
    'Undisclosed investors':
    'Undisclosed',
    'Undisclosed investor': 'Undisclosed',
    'Not disclosed': 'Undisclosed',
    'Undisclosed': 'Undisclosed'
})

[19]: #1 WORD USE FOR MANY ex., 'Seed Funding' = 'Seed Funding', 'Seed', 'Seed Round'

df['round'] = df['round'].str.strip()

round_mapping = {
    'Seed Funding': 'Seed Funding',
    'Seed': 'Seed Funding',
    'Seed Round': 'Seed Funding',
    'Seed/ Angel Funding': 'Seed/Angel Funding',
    'Seed / Angel Funding': 'Seed/Angel Funding',
    'Seed/Angel Funding': 'Seed/Angel Funding',
    'Angel': 'Seed/Angel Funding',
    'Angel / Seed Funding': 'Seed/Angel Funding',
    'Private Equity': 'Private Equity',
    'Debt Financing': 'Debt Funding',
    'Debt Funding': 'Debt Funding'
}

[21]: df.head()

```

• Data Cleaning Output – Step 2 (Cleaned Table)

After applying all preprocessing steps, the final cleaned dataset was generated. This included properly formatted numerical columns, corrected text values, and consistent records across all fields. The cleaned dataset was displayed again to validate that the transformations were applied successfully.

This output confirms the dataset is now accurate, consistent, and ready for visualization and machine learning implementation.

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JupyterLab Python [conda env:base]

```
[56]: df.head(10)
```

	startup	industry	city	investors	round	amount crores	year	state
0	Fame	Media	Mumbai	To the new ventures	Private Equity	82.50	2015	Maharashtra
1	1mg	Healthcare	Gurgaon	Deep kalra, Sequoia capital, Omidyar network, ...	Private Equity	49.50	2015	Haryana
2	20Dresses	Personal Care	Mumbai	Srinivas aluri, Srinivas modukury, Bs nagesh	Seed Funding	8.25	2015	Maharashtra
3	33Coupons	Marketing	Kanpur	Undisclosed	Seed Funding	0.17	2015	Uttar Pradesh
6	Aasaanjobs	Careers	Mumbai	Inventus capital partners, Idg ventures	Seed Funding	12.38	2015	Maharashtra
8	Absentia	Technology	Banglore	Astarc ventures, 50k ventures & others	Seed Funding	1.49	2015	Karnataka
10	AdStringO	Marketing	Mumbai	Arihant patni, Vishal khare, viswanath ramach...	Seed Funding	2.89	2015	Maharashtra
11	AdWyze	Marketing	Banglore	Mape advisory	Seed Funding	8.25	2015	Karnataka
12	AgentDesks	Real Estate	Gurgaon	Sierra ventures, Cota capital, Vegas tech fund	Seed Funding	24.34	2015	Haryana
13	Agrostar	AgriTech	Pune	Idg ventures	Private Equity	33.00	2015	Maharashtra

• Machine Learning Output (Linear Regression)

A Linear Regression model was applied to understand and predict trends in startup funding amounts based on historical data. The focus of the ML execution was to analyze numerical relationships rather than build a complex prediction interface.

The model output consisted of:

Regression coefficients

Intercept value

Predicted output based on sample inputs

Since no predictive graph was generated, the printed output served as the model execution result. This demonstrates that the ML model successfully ran and produced meaningful numerical outputs.

```
103.0000/notebooks/ind_funding.ipynb:
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JupyterLab Python [conda env:base]

[285]: # TOTAL FUNDING PREDICTION BY YEAR (LINEAR REGRESSION)

yearly = df.groupby('year')['amount crores'].sum().reset_index()
X = yearly['year'].values.reshape(-1,1)
y = yearly['amount crores'].values
model = LinearRegression()
model.fit(X, y)
future_years = np.array([2024]).reshape(-1,1)
predictions = model.predict(future_years)
np.set_printoptions(precision=2, suppress=True)
print(predictions)

[197566.12]

[286]: # STARTUPS COUNT PREDICTION BY YEAR (LINEAR REGRESSION)

count_year = df['year'].value_counts().sort_index().reset_index()
count_year.columns = ['year', 'startup_count']
X = count_year[['year']]
y = count_year['startup_count']
model = LinearRegression()
model.fit(X, y)
future_years = pd.DataFrame({'year': [2024]})
future_startups = model.predict(future_years)
future_startups_rounded = future_startups.round().astype(int)
future_startups_rounded

[286]: array([268])
```

5.2 Test Plan

The test plan outlines the strategy and methodology adopted to validate the integrity and functionality of the project. The purpose of the test plan is to confirm that the data is accurate, the system behaves as expected, and the dashboard responds correctly to various user interactions.

The following aspects were included in the test plan:

Verifying correct dataset loading

Confirming data cleaning steps

Testing the functionality of slicers in the dashboard

Ensuring KPI metrics update correctly

Validating ML model outputs for correctness

The test plan ensures that the entire system—from data processing to visualization—works without errors.

5.3 Data Validation Test Cases

Data validation test cases were performed to check whether the dataset meets the required standards and is suitable for further analysis.

The following data validation checks were carried out:

Null Value Verification: Ensured no missing data remains after cleaning

Duplicate Check: Confirmed removal of repeated entries

Data Type Validation: Checked that numerical fields are stored as numbers

Outlier Detection: Identified unusually high or low values

Range Validation: Ensured funding years fall within a valid range

These validation outputs ensure the dataset is consistent, clean, and reliable.

5.4 Functional Validation Test Cases & Results

Functional testing was performed to ensure the entire analytical system behaves correctly under various user actions. This includes testing dashboard interactions, slicer functionality, and the correctness of KPIs and charts.

Key functional test scenarios include:

Selecting different years in slicers updates all visuals instantly

Changing city filter updates city-wise funding charts

Selecting industry modifies related charts and KPIs

KPIs such as Total Funding and Average Funding update accurately

ML model runs without errors and displays output correctly

These tests confirm that both the Power BI dashboard and Jupyter Notebook components function as intended, providing a smooth and accurate analytical experience.

6. Conclusion and Recommendations

• Conclusion

The analysis of startup funding data provided a deep and meaningful understanding of how investments are distributed across different years, cities, and industries. By combining the strengths of Python-based data processing with the intuitive visualization capabilities of Power BI, this project transformed a raw dataset into an insightful decision-support system.

Through a structured workflow—beginning with data import, cleaning, validation, visual exploration, and concluding with a predictive model—the project successfully demonstrated the power of data analytics in uncovering patterns that are often overlooked. The interactive Power BI dashboards made it possible to explore funding behavior dynamically, while the histogram and machine learning outputs from Jupyter Notebook enriched the analysis with statistical depth.

The findings reveal that startup funding in India is concentrated in major metropolitan regions and key industries, particularly technology-driven sectors. Year-wise trends also show fluctuations in investment volume, influenced by economic conditions, market confidence, and ecosystem maturity. The Linear Regression model, though simple, confirmed observable trends and laid the foundation for predictive analysis.

Overall, the project achieves its core objective: to provide analysts, investors, and stakeholders with a clear, data-backed understanding of the startup funding landscape. The integration of analytics, visualization, and machine learning showcases how data-driven decision-making can guide strategies, identify opportunities, and support better planning in a highly competitive market.

• Recommendations

Based on the insights gained and the performance of the analytical system, several recommendations can be proposed to enhance the project further and unlock even deeper value:

1. Expand the Dataset for Deeper Insights

Including more years, additional startup features (revenue, valuation, employee count), and investor-related data can create richer patterns and improve the accuracy of predictions.

2. Adopt Advanced Machine Learning Models

While Linear Regression provides a basic trend analysis, more advanced models—such as Random Forest, XGBoost, or Time-Series Forecasting—can capture complex patterns and offer stronger predictive capabilities.

3. Build a Web-Based or Cloud Dashboard

Deploying the Power BI report online (Power BI Service/Embedded) will allow investors, entrepreneurs, and analysts to access insights anytime, making the system more practical and scalable.

4. Add More Interactive Features in Power BI

Enhancing the dashboard with drillthrough pages, detailed tooltips, comparative charts, and industry-wise trend pages will provide users with a richer analytical experience.

5. Enable Automated Data Refresh

With scheduled data updates or API-based integration, the dashboard can evolve into a real-time funding tracker—valuable for investors and researchers monitoring the startup ecosystem.

6. Improve Data Quality with Advanced Cleaning Techniques

Standardizing values more deeply, removing outliers statistically, and applying NLP for text fields can increase accuracy and enhance model outcomes.

7. Integrate Real-Time Funding APIs

Connecting the system with platforms like Startup India, Tracxn, or Crunchbase will ensure the dashboard always reflects current market conditions.

8. Add More KPIs and Business Metrics

Metrics like median funding, top industries by CAGR, investor activity score, or city-wise average deal size can bring more business value to the dashboard

7. Future Scope

The Startup Funding Analysis system has strong foundations in data cleaning, visualization, and basic prediction. However, it can be enhanced further to deliver more intelligent and industry-ready insights. The major future improvements include:

- **Integration of Advanced ML Models**

Beyond linear regression, models like Random Forest, XGBoost, and Time-Series Forecasting can be added to provide more accurate predictions of future funding trends and investor behavior.

- **Real-Time Data Automation**

The system can be upgraded to fetch live startup funding data using APIs or web scraping. This will make the dashboards dynamic and always up to date.

- **Interactive Web Application**

Power BI dashboards can be transformed into a complete web-based data analytics platform using Streamlit or Power BI Embedded, allowing easy access to users anywhere.

- **Smart Insights Using NLP**

Natural Language Processing can be used to generate automatic insights, detect emerging industries, and summarize funding news.

- **Recommendation Engine for Startups**

The system can be extended to recommend the best investors, locations, and funding rounds for new startups based on historical success patterns.

8. Bibliography & References

Books & Academic Sources

- Wes McKinney, Python for Data Analysis, O'Reilly Media.
- Jake VanderPlas, Data Science Handbook, O'Reilly Media.
- Alberto Cairo, The Truthful Art: Data, Charts, and Maps, New Riders.

Web Resources

- Pandas Documentation – <https://pandas.pydata.org> □
- NumPy Documentation – <https://numpy.org> □
- Matplotlib Documentation – <https://matplotlib.org> □
- Scikit-Learn Documentation – <https://scikit-learn.org> □
- Power BI Official Documentation – <https://learn.microsoft.com/power-bi> □
- Kaggle Dataset Reference (If used) – <https://kaggle.com> □
- Medium Articles on Data Analysis & Visualization – <https://medium.com> □
- Tools & Software
- Jupyter Notebook – <https://jupyter.org> □
- Power BI Desktop – Microsoft Power BI
- Python 3.12 – Python Software Foundation